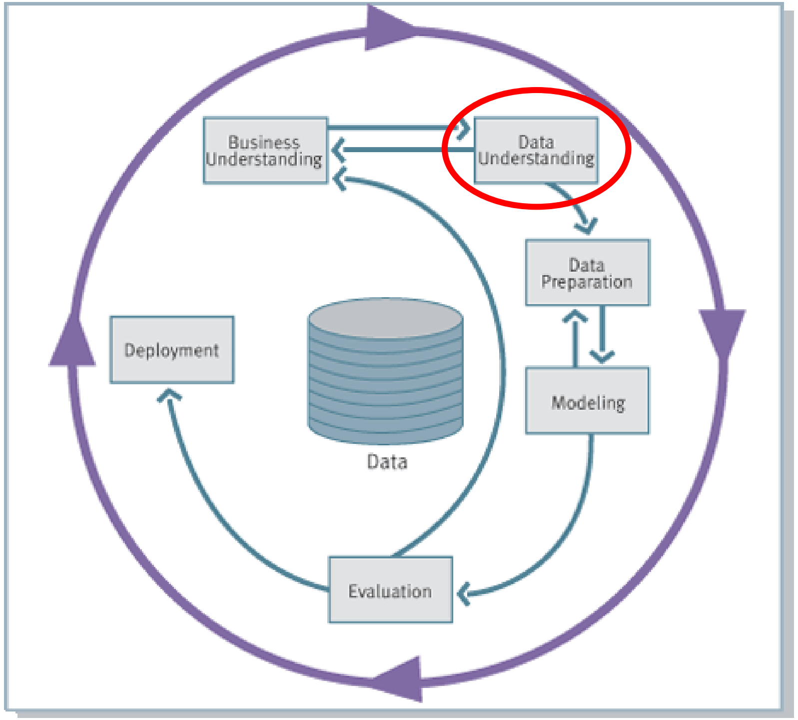
Data Understanding



Data Understanding

Collect Initial Data

The first task in data understanding is to access relevant data in the inventory of resources.

Example: load data from a data warehouse

OUTPUT:

- Initial data collection report, which list the data with locations, methods used to acquire them, and any problems encountered.

Data Understanding

Describe Data

It's the first time in which the properties of the data are examined.

OUTPUT:

- Data description report, which describes data in terms of format, potential values, quantity, identities of the fields, and any characteristics which is discovered.

Metadata collected in a datawarehouse can help.

Data Understanding

Describe Data

There are two major types of variables:

- **Categorical:** The possible values are finite and differ in kind. There are two subtypes:
    - ***nominal*** variables name the kind of object to which they refer, but there is no order among the possible values.
        - *Marital status*: single, married, divorced, unknown
        - *Gender*: male, female
        - *Level of education*: university, college, high school.
    - ***ordinal*** variables have an order among the possible values.
        - *Customer credit rating*: good, regular, poor

Data Understanding

Describe Data

- **Quantitative:** Arithmetic operations are allowed and sensible. There are two subtypes:
  - ***discrete*** variables, whose values are integers.
    - *Number of employees*
    - *Time of year (month, season, quarter)*
  - ***continuous*** variables, whose values are real numbers.
    - *Income*
    - *Average number of purchases*
    - *Revenue*

Data Understanding

Describe Data

#	Patient ID	Blood type	Pain level	Number of visits	Body temperature (°C)	Hospitalized?
1	P-001	A+	Mild	3	36.7	No
2	P-002	O−	Severe	12	38.4	Yes
3	P-003	B+	Moderate	1	37.1	No
4	P-004	AB+	None	7	36.2	No
5	P-005	A−	Mild	2	37.8	No
6	P-006	O+	Extreme	20	39.1	Yes
7	P-007	B−	Moderate	5	36.9	No

Data Understanding

Verify Data Quality

Data quality is inspected, addressing the following issues:

- Accuracy: conformity of stored value to the actual value.
- Completeness: no missing values.
- Consistency: uniform representation.
- Up-to-dateness: stored data is not obsolete.

Poor data quality and ***poor data integrity*** are major issues in almost all KDD projects.

Data Understanding

Verify Data Quality

Most operational data has never been captured or modeled for data mining purposes.

The selected data is typically collected from numerous, inconsistent, poorly documented operational systems.

Even where the data is augmented with ***external*** data, trying to match internal and external records can be a major challenge.

Purchased databases often offers up to 400 variables, which are usually very sparsely filled in and can be out of date.

Data Understanding

Verify Data Quality

It is important to understand the time sensitivity of data.

Example: a percentage of customers will change their jobs every year: any analysis where job type is a factor has to be re-examined periodically.

The data management specialist is responsible for collecting and integrating the data into the informational environment.

Data Understanding

Verify Data Quality

OUTPUT:

- Data quality report, which reports findings of the data quality verification. If quality problems exists, it also discusses possible solutions.

Data Understanding

Explore Data

Finally data exploration starts.

How?

statistical methods

+

data visualization techniques

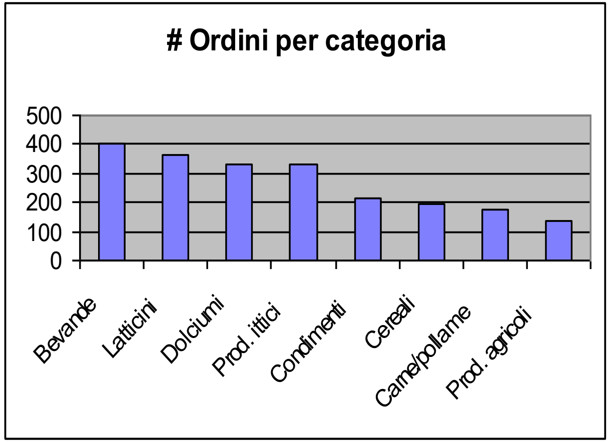
OUTPUT:

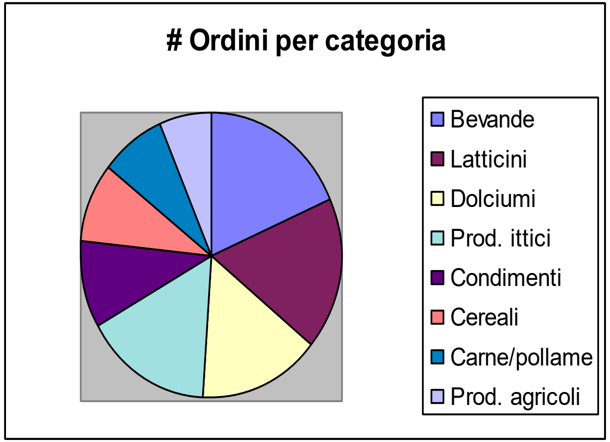
- a data exploration report

Data Understanding

Explore Data

For *categorical variables*, frequency distributions of the values are a useful way of better understanding the data content. Histograms and pie charts help to identify distribution skews and invalid or missing values.





Data Understanding

Explore Data

When dealing with *quantitative variables*, the data analyst is interested in such measures as

- maximum and minimum,
- mean (average),
- mode (most frequently occurring value),
- median (midpoint value),
- statistical measures for central tendency (standard deviation).

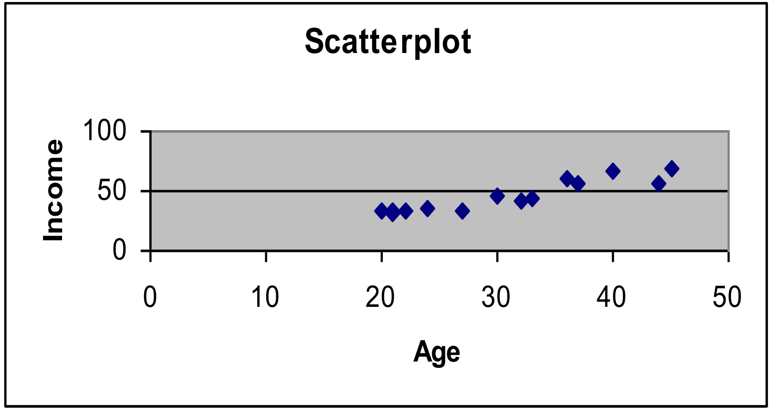
When combined, these measures offer a powerful way of determining the presence of invalid and skewed data.

<u>Example</u>: maxima and minima analysis quickly show up spurious values.

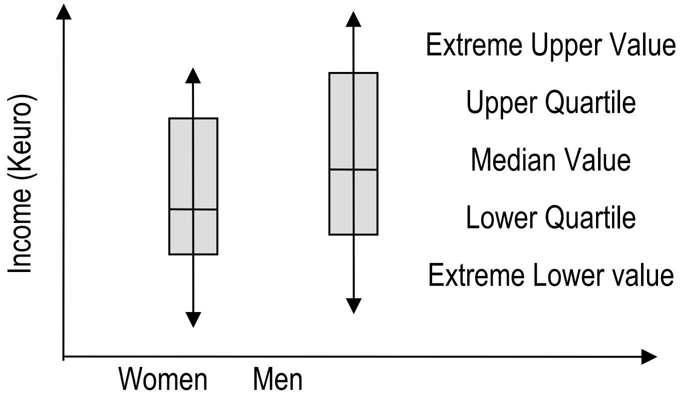
Data Understanding

Explore Data

Other graphical tools are scatterplots ad boxplots.



Two-dimensional pictures that represent the relationship between two or more continuous variables



Useful for comparing the center (average) or spread (deviation) of two or more variables.

Customer Vulnerability Analysis (CVA)

The main data source a syndicated database of some 15,000 consumer whose supermarket purchases had been tracked for three years. Tracking was possible thanks to “fidelity cards”.

For each supermarket visit, the following data are available:

- Customer ID
- Date of visit
- List of items purchased (the ***basket***)
- Product quantity for each item
- Promotion ID (whether the item was bought at full price, or through some promotion such as coupons).
- Basket value

Customer Vulnerability Analysis (CVA)

More precisely, data are stored into three tables of a relational database:

- The table Visits with the following structure:
    - Visit ID (primary key)
    - Customer ID (foreign key)
    - Date of visit
    - Basket value
    - Payment mode

Customer Vulnerability Analysis (CVA)

- The table Purchases with the following structure:
  - Purchase ID (primary key)
  - Visit ID (foreign key)
  - Product ID (foreign key)
  - Product quantity
  - Promotion ID
- The table Products with the following structure:
  - Product ID (primary key)
  - Description
  - Quantity
  - Brand

Customer Vulnerability Analysis (CVA)

By inspecting the nature of data the following conclusions are drawn:

- Table Visits:
  - Visit ID: nominal
  - Customer ID: nominal
  - Date of visit: date
  - Basket value: continuous
  - Payment mode: nominal

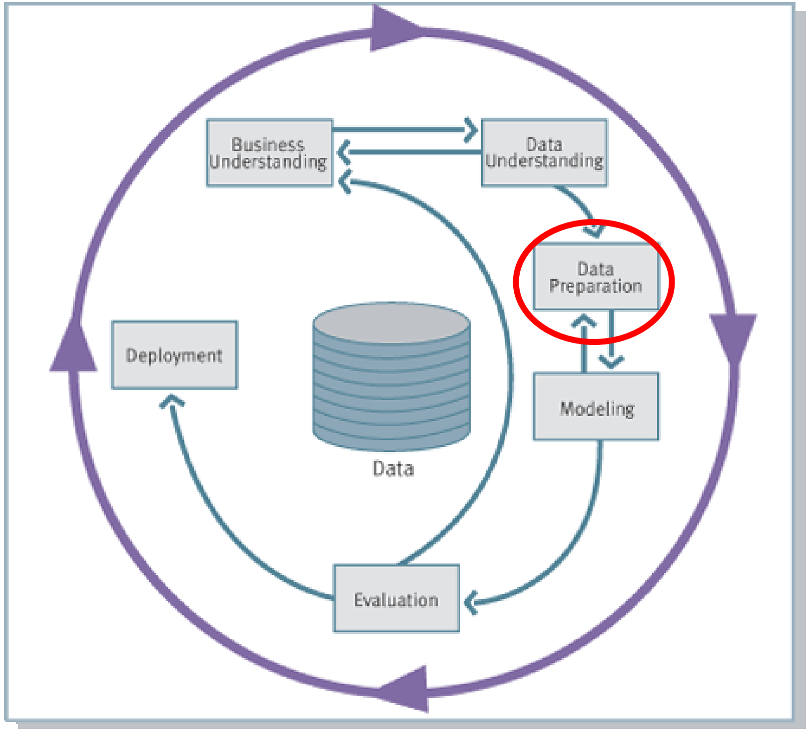
Customer Vulnerability Analysis (CVA)

- Table Purchases:
  - Purchase ID: nominal
  - Product ID: nominal
  - Visit ID: nominal
  - Product quantity: discrete
  - Promotion ID: nominal
- Table Products:
  - Product ID: nominal
  - Description: nominal (free text)
  - Quantity: either discrete (e.g., two rolls) or continuous (e.g., 0.33 cc)
  - Brand: nominal

Customer Vulnerability Analysis (CVA)

- Data are generally of high quality, but in several records of Purchases the Product ID does not properly reference a product in Products.
- For these cases, a partial view of the basket is built.
- The description of a product is free text. However, the text is not relevant for the analysis but only for the interpretation of final patterns.

Data Preparation



Data Preparation

Select Data

Data selection concerns

- Tuples of database relations

Selection criteria include:

- Relevance to the data mining goals
- Quality and technical constraints
- Limits on data volumes or data types.

In an age when storing terabytes of data is common, the importance of reducing the size of a data set increases.

OUTPUT:

- report with a rationale for inclusion/exclusion

Data Preparation

Select Data

Data selection can be performed

- Manually
- Automatically → sampling & feature selection

Select Data

Sampling

- Simple random sampling is the most straightforward type of sampling: **every** conceivable **group of objects** of the required size **has *the same chance* of being the selected sample**.
- It is possible to get a very atypical sample, however the laws of probability dictate that *the larger a sample is, the more likely it will be representative of the population whence it came*.

Select Data

Sampling

- The traditional method of choosing a random sample begins with a numbering of the members of the target population. The order in which they are numbered is irrelevant. Once each member in the population has a number, the sampler consults a random number table to select the indices of the members to be included in the sample.
- *Main assumption*: we are able to number the members of the target population.

Select Data

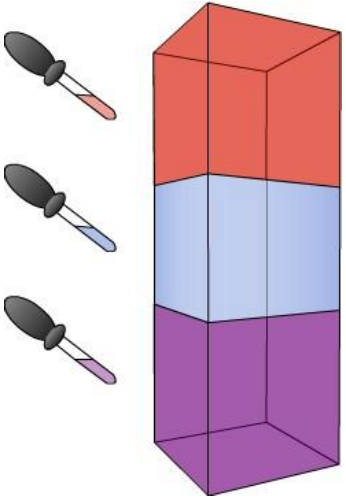
Sampling

- Two types of simple random sampling:
  - With replacement
  - Without replacement
- If the population is large compared to the sample, there is a very small probability that any member will be chosen more than once and the two techniques are essentially the same.

Select Data

Sampling

- Stratified random sampling: useful when the target population is divided into strata, or groups. In stratified random sampling a simple random sample is then selected from each "stratum" separately, and their union produces a stratified sample.



Select Data

Sampling

The percentage of samples from each stratum depends on the goal of stratified sampling:

- If you want to ensure that each subgroup, or stratum, within the population is represented adequately in the sample, by reflecting its proportion in the overall population, then the percentage of samples from each stratum corresponds to the relative size of the stratum.

Select Data

Sampling

If the target population is divided into three strata, of size 5%, 30% and 65%, then a simple random sample of 1% within each group produces a stratified sample composed of 0.05%, 0.30% and 0.65% of cases from the three strata respectively.

- In this example, the analyst ensures that the sample includes representatives of all strata. Indeed, when one stratum has a much smaller membership than the others, simple random sampling **from the whole dataset** can produce samples without representative of that stratum.

Select Data

Sampling

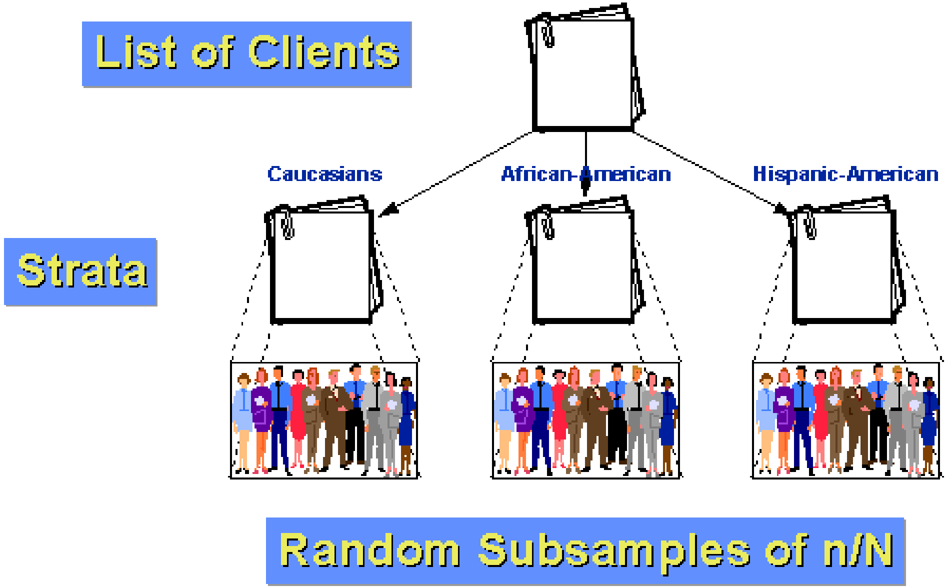
- On the other hand, you may want to control the number of observations within each stratum or group in order to ensure that particular groups within a population are adequately represented in the sample.
- For instance, if the number of samples for each stratum is 25 in order to get something meaningful, then you may decide that you will draw at least 25 samples from each stratum, **independently of the relative size**.

Select Data

Sampling

Example: a population of 1000 customers can be divided into three groups: Caucasian, African-American and Hispanic-American.

The last two groups are relatively small minorities (10% and 5% of the population, respectively).



Select Data

Sampling

Example (cont.): If we just did a stratified random sample of n=100 by respecting the original size of each subgroup, then we would have a sampling of 85 Caucasians, only 10 Afro-Americans and just 5 Hispanic-Americans.

But we think that in order to say anything about subgroups we will need at least 25 cases in each group!

So, we sample 50 Caucasians, 25 African-Americans, and 25 Hispanic-Americans. Therefore, we have a within-stratum sampling fraction of

- 25/100 = 25% for African-Americans
- be 25/50 = 50% for Hispanic-Americans
- 50/850 = about 5.88% for Caucasian customers

Select Data

Sampling

Advantages:

- If the members within strata are more similar to each other than members from different strata, stratum-specific estimates will be more precise than ones from the entire sample.

  In the previous example, we know we will have enough cases from each group to make meaningful **subgroup** inferences because of stratification.

Warning: it is important to adjust for the overrepresentation when inferences (generalizations) refer to the target population as a whole (i.e., across the strata).

Select Data

Sampling

If we estimate the average income from a stratified sampling of 85 Caucasians, 10 Afro-Americans, and 5 Hispanic-Americans, then the statistics would not be a biased estimation of the average income on the whole population of 1000 samples.

However, if we estimate the average income from a stratified sampling of 50 Caucasians, 25 Afro-Americans, and 25 Hispanic-Americans, then the contribution of the minorities would be overemphasized. We have to correct it by weighting the average of each group:

Avg_income =

0.85*Avg_income_Caucasian + 0.1*Avg_income_Afro-American + 0.05*Avg_income_Hispanic-Americans.

Select Data

Sampling

What are the rules of a good design for a stratified sampling procedure?

- Strata should be chosen to:
  - have means which differ substantially from one another
  - minimize variance within strata and maximize variance between strata.
- Sample sizes should be made proportional to the stratum standard deviation (the larger the standard deviation, the greater the sample size), unless we need to control the number of observations within each stratum, as seen earlier.

Select Data

Sampling

- Cluster Sampling. Sometimes members in the population naturally form clusters.

  Examples: customers' purchases can be clustered along time (by week, month, or quarter). Customers can be clustered by living area.

  Then it is possible to sample clusters.

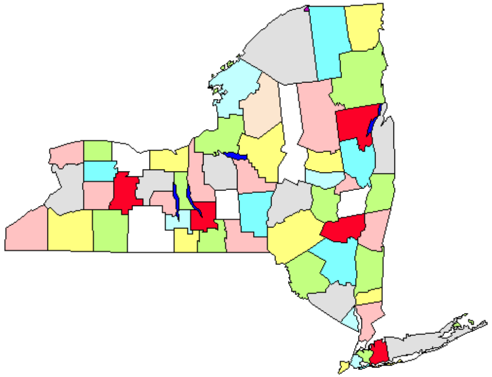
  Examples: selecting respondents from certain areas only, or certain time-periods only.

  In this case, all members of each sampled clusters are considered.

Select Data

Sampling

Example: consider the map of the counties in New York State. Assume you have to do a survey of town governments that require you going to the towns personally. If you do a simple random sample state-wide you have to cover the entire state geographically. Instead, you decide to do a cluster sampling of five counties (marked in red). Once these are selected, you go to every town government in the five areas. Clearly this strategy will help you to economize on your mileage.



Select Data

Sampling

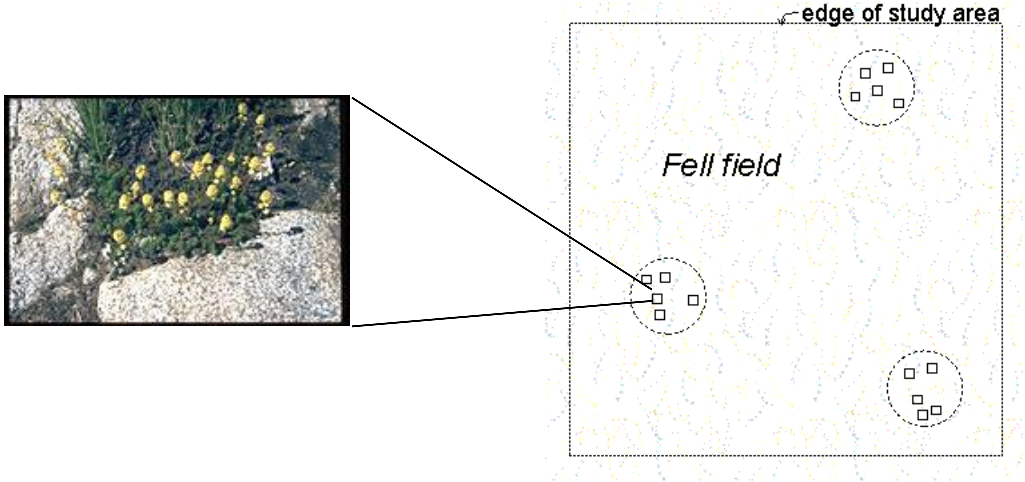
- In the context of large databases, a common application of cluster sampling is to randomly select blocks of data, and then use all data on these blocks. The motivation behind this approach is that to retrieve one database record from a particular block, the entire block must be read into memory. This type of sampling is also called block sampling.

S. Chaudhuri, R. Motwani & V. Narasayya (1998). Random sampling for histogram construction: how much is enough. Proceedings of ACM SIGMOD Conference, pp. 436-447.

Select Data

Sampling

- Two-stage sampling. Combines two ideas: randomly choosing clusters, and then sampling within each cluster.



Select Data

Sampling

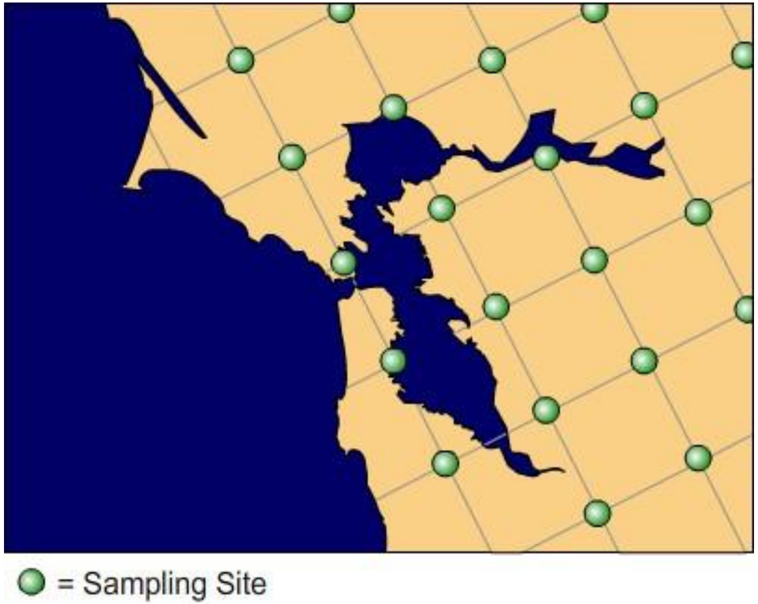
- Systematic sampling. When individuals in a population are numbered in some way, systematic sampling is an option. Also known as every k–th sampling, systematic sampling involves choosing one member at random from those numbered between 1 and k, then including every k–th member after this in the sample.

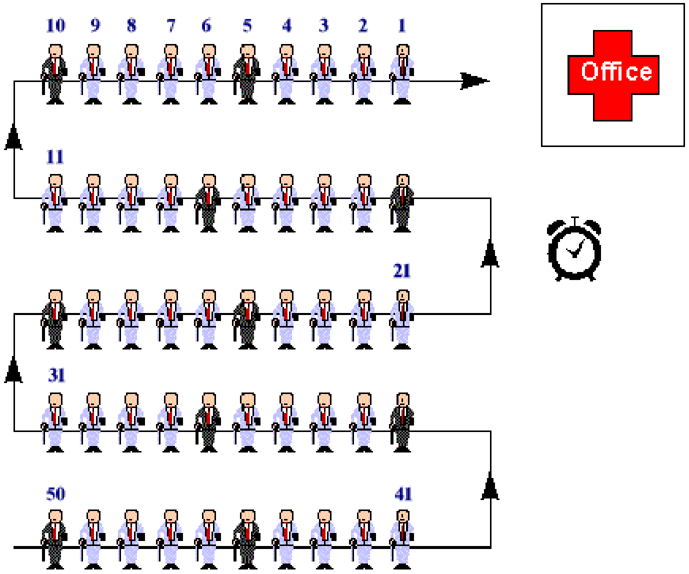
  Example: selecting every 10–th name from the telephone directory.

Select Data

Sampling

Other two examples:





Select Data

Sampling

Main *advantage*:

easy to implement, even in the case in which the actual size of the population is initially unknown or counting the members of the whole population is computationally expensive.

Warning: Because the selection is not random, systematic samples may not be representative of the population and must be used carefully. It is especially vulnerable to periodicities in the list of members of the population. If periodicity is present and the period is a multiple of *k*, then bias will result.

Select Data

Sampling

How large should the sample be?

There are straightforward mechanisms in statistics for estimating the sample size necessary to have a certain probability of detecting an effect of a pre-specified size or greater. These tools, grouped under the heading of power analysis, are based on estimates of means and variances of the variables in the population to be sampled.

Select Data

Sampling

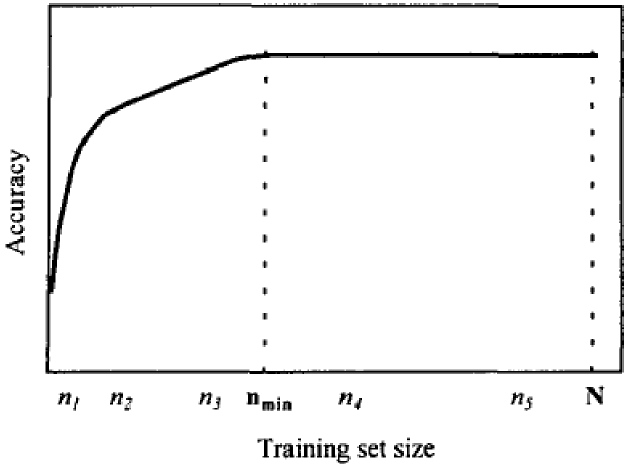
It is important to understand the cause system of which the population are outcomes and to ensure that all sources of variation are considered. *Large number of observations are of no value if major sources of variation are neglected in the study.*

An alternative to setting a predetermined sample size is to let the data choose the size of the sample. The basic idea is to continue to increase the sample size until the results or summaries no longer change very much (where "very much" is set in advance). Variations on this idea are known as progressive sampling, adaptive sampling or sequential sampling.

Select Data

Sampling

Converging learning curve



A generic algorithm that defines a family of progressive sampling methods.

Compute schedule $S = \{n_0, n_1, n_2, \ldots, n_k\}$ of sample sizes
$n \leftarrow n_0$
$M \leftarrow$ model induced from n instances
while not *converged*
    recompute S if necessary
    $n \leftarrow$ next element of S larger than n
    $M \leftarrow$ model induced from n instances
end while
return M

F. Provost, D. Jensen, T. Oates (1999). Efficient progressive sampling. *Proceedings of the Fifth International Conference on Knowledge Discovery and Data Mining*, pp. 23 – 32.

Select Data

Feature selection

- Real-life databases contain many attributes (also called features)
- The problem of feature selection arises because
  - search complexity in the hypothesis space must be reduced for practical reasons and
  - redundant or irrelevant features can have significant effects on the quality of results of the analysis method (the curse of dimensionality)

Select Data

Feature selection

#	Blood type	Pain level	Visits	Temp (°C)	High fever ?	Frequent visitor?	Pain score (0–4)	Day of week	Zodiac sign	Hospitalized?
P-001	A+	Mild	3	36.7	No	No	1	Mon	Virgo	No
P-002	O−	Severe	12	38.4	Yes	Yes	3	Fri	Aries	Yes
P-003	B+	Moderate	1	37.1	No	No	2	Wed	Capricorn	No
P-004	AB+	None	7	36.2	No	Yes	0	Tue	Scorpio	No
P-005	A−	Mild	2	37.8	Yes	No	1	Sun	Gemini	No
P-006	O+	Extreme	20	39.1	Yes	Yes	4	Thu	Leo	Yes
P-007	B−	Moderate	5	36.9	No	No	2	Sat	Libra	No

Feature Selection

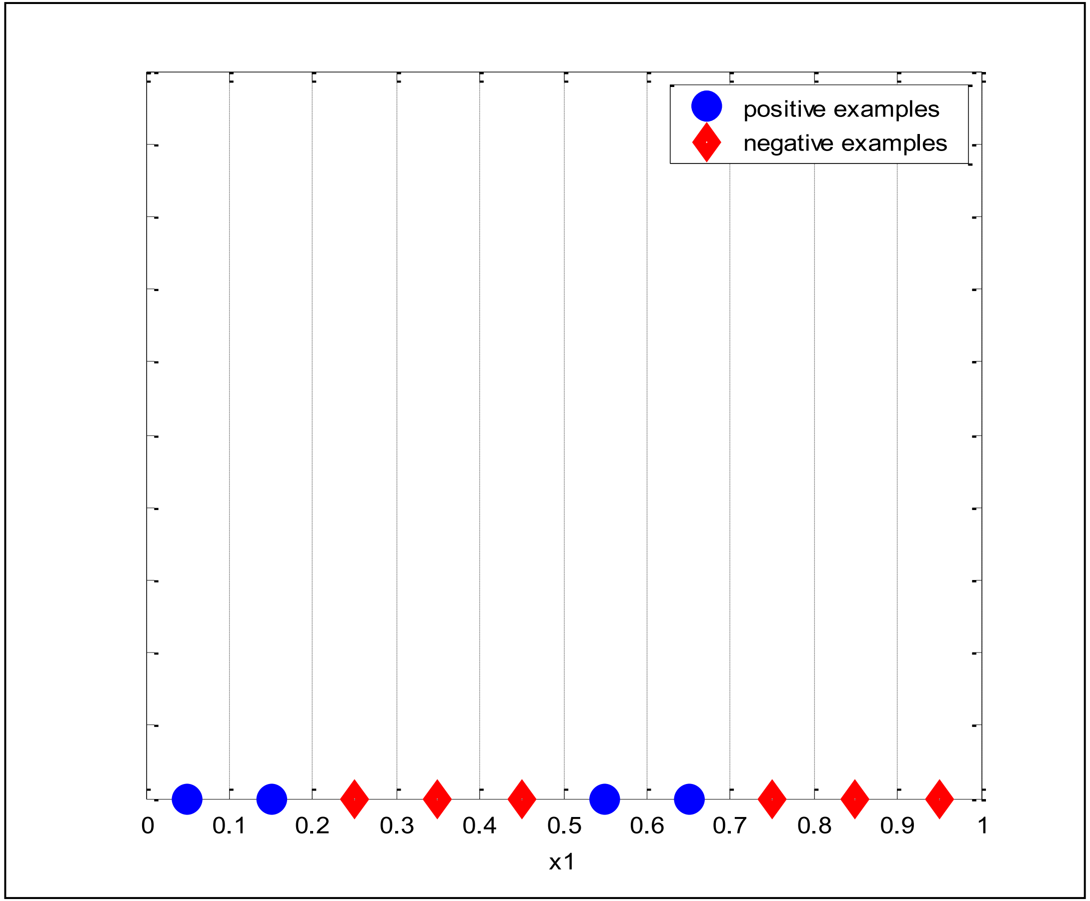
The curse of dimensionality

The basic idea of the curse of dimensionality is that high dimensional data is difficult to work with for several reasons:

- The required number of samples (to achieve the same accuracy) grows exponentially with the number of variables! In practice the number of training examples is fixed-> *the classifier's performance usually will degrade for a large number of features*.
- There aren't enough observations to get good estimates.

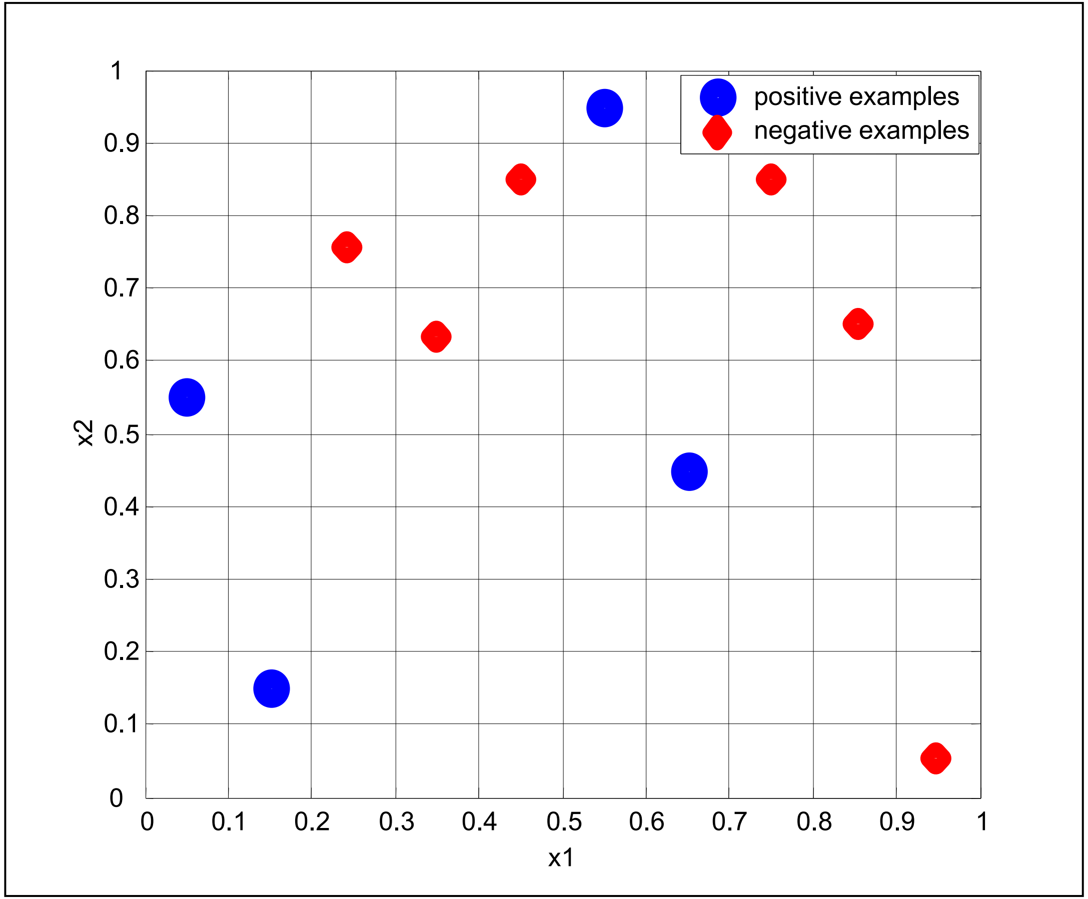
Feature Selection

The curse of dimensionality



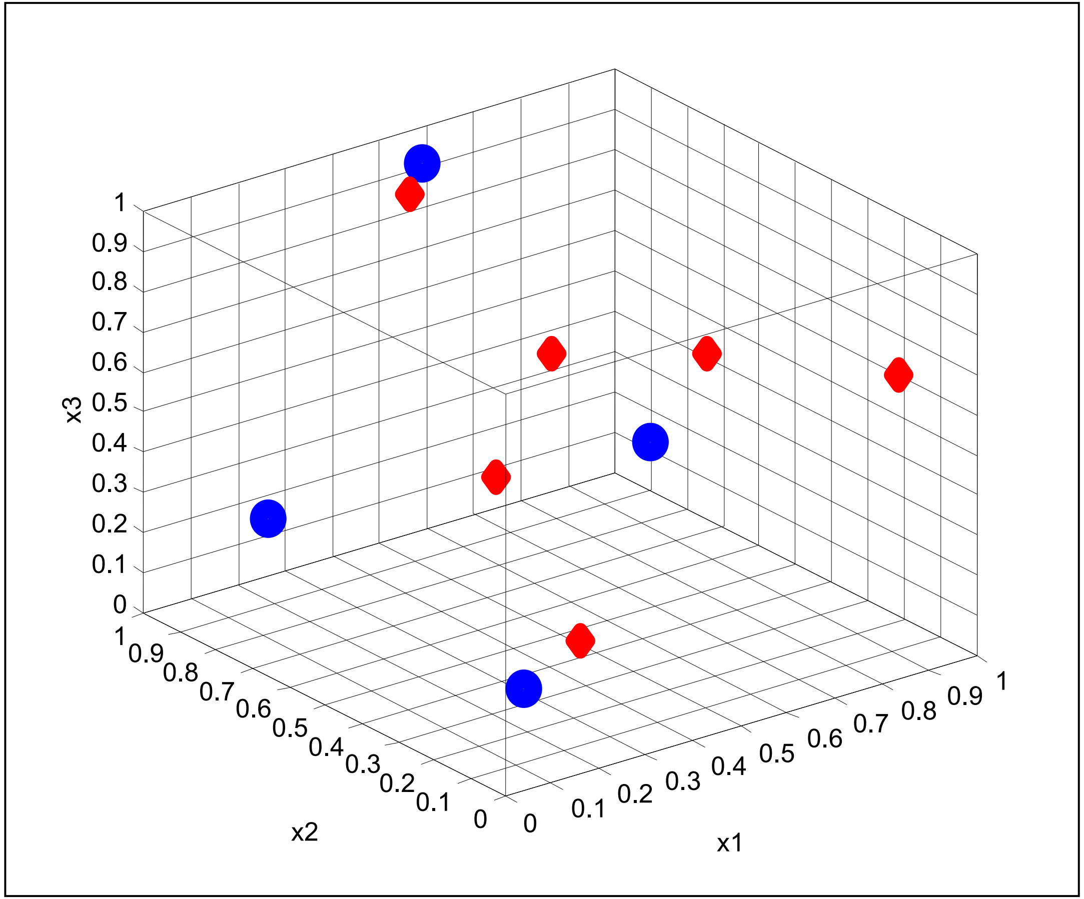
Feature Selection

The curse of dimensionality



Feature Selection

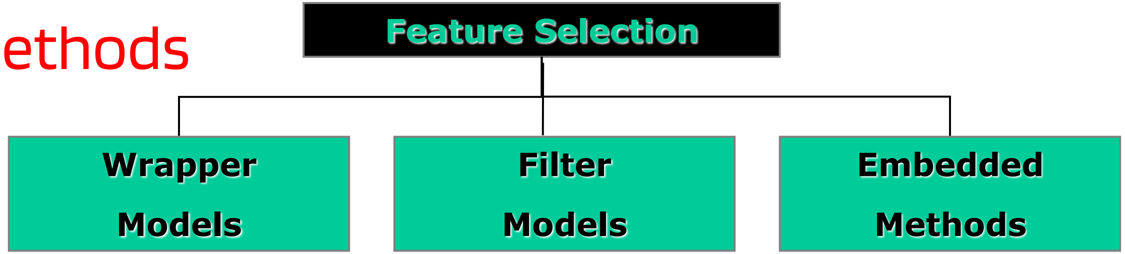
The curse of dimensionality



Select Data

Feature selection

- Feature selection is a process that chooses an optimal subset of features according to a certain criterion.
- Three approaches:
    - Wrapper models
    - Filter models
    - Embedded methods



Select Data

Feature selection

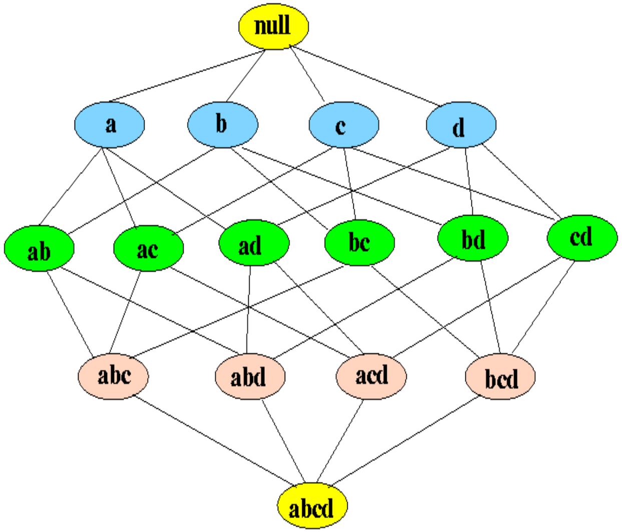
- The **wrapper model** relies on a data mining algorithm to determine whether a subset of features is good.
- The algorithm is used as a part of evaluation function and also to induce the final model or patterns.
- The DM algorithm can solve either a predictive (e.g., classification, regression) task or a descriptive (e.g., clustering) task.

Select Data

Feature selection

- The process of feature selection in the wrapper model involves three main steps:

1. Feature Subset Generation: The wrapper model generates multiple subsets of features by selecting a subset of features from the original feature set. The generation of the subsets is normally based on some search strategy that explores only a part of the feature space, which is the lattice of subsets:



Feature Selection

Feature subset selection strategies

Enumeration of all possible subsets and selection of the best.

Random generation of subsets and selection of the best.

Sequential generation of subsets.

- Forward selection: start with an empty subset and gradually add one feature at a time.
- Backward selection (or elimination): start with a full set and remove one feature at a time.

These strategies are based on a greedy search strategy in the space of 2^N subsets, where N is the number of features.

Select Data

Feature selection

2. Model Training: The generated subsets are used to train a model, and the model's performance is evaluated using a performance metric such as accuracy or precision.

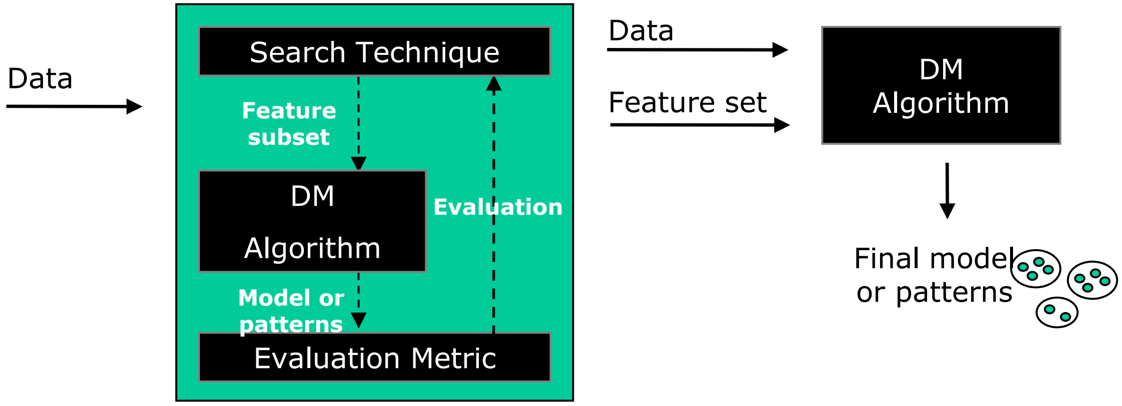
How to evaluate a performance metric in an unbiased way?

- Holdout method: The data is split into two sets: a training set and a testing set. The model is trained on the training set and then evaluated on the testing set. Advantage: easy to implement. Disadvantage: it may result in a biased estimate of the model's performance if the random split generate (by chance) a biased partition.
- Cross-validation: It involves dividing the data into *k*-folds, where *k* is usually set to 5 or 10. The model is trained on *k*-1 folds and tested on the remaining fold. This process is repeated *k* times, with each fold being used as the testing set once. The results are then averaged to obtain an estimate of the model's performance.

Select Data

Feature selection

3. Subset Evaluation: The subsets are evaluated based on the model's performance, and the subset that results in the best performance is selected.



Select Data

Feature selection

Stopping criteria. In any case we should ensure that the loop of generation and evaluation terminates.

For exhaustive or sequential feature generation, the loop will naturally stop when a full set becomes empty, or an empty subset becomes full.

In other cases, we need to force the loop to terminate, such as

- when a subset is good enough with respect to the performance measure or
- when a certain percentage of the total search space has been searched (a variation of this criterion is the number of subsets that have been generated).

Select Data

Feature selection

Rationale for a wrapper approach. If we want to have a good set of features to improve the accuracy of a learning algorithm, why don't we use the performance of the learning algorithm as a measure?

The wrapper model differs from other feature selection methods in that it uses the performance of a model trained on a subset of features to evaluate the importance of those features. This allows the wrapper model to select the best features for a specific model and problem domain.

Select Data

Feature selection

Issues:

- How to truly estimate predictive accuracy and avoid overfitting?
  - Hold-out / cross validation

- Training a model takes time, so the wrapper model is computationally expensive, as it requires training multiple models on different subsets of features.
  - However, it can provide better results than other feature selection methods by selecting the most relevant features for a specific model and problem domain

Select Data

Feature selection

- The **filter model** is independent of the data mining algorithm that will use the subset of features.
- Also for filter models, the generation of the subsets is normally based on some search strategy that explores only a part of the feature space.
- No matter which method of feature subset generation is adopted, we need some measure to decide which feature should be added or removed, or which subset should be kept.

Feature Selection

Feature evaluation

- Subsets are evaluated by using some measures of intrinsic properties of the data.
- We will consider the case of classification tasks (samples belong to one class).
- In this case two families of measures are:
  - Information measures
  - Distance measures

Feature Selection

Feature evaluation

Information measures

- For example: information gain (IG) measures how much our uncertainty about classes decreases after observing a feature X. (we will see this measure with decision trees)
- It is calculated as the difference between the initial uncertainty about the classes (calculated using prior probabilities) and the average remaining uncertainty after observing the different possible values of feature X (using conditional probabilities).

Feature Selection

Feature evaluation

Distance measures

These measures allow us to quantify the relevance of features by evaluating how much the patterns of different classes are separated in the feature space, rather than relying on uncertainty reduction.

- Example: Consider two features for classifying fruits: weight and diameter. A distance measure can reveal that diameter separates apples from oranges better than weight, since the clusters of the two classes are more distant when projected onto the diameter axis.:

Select Data

Feature selection

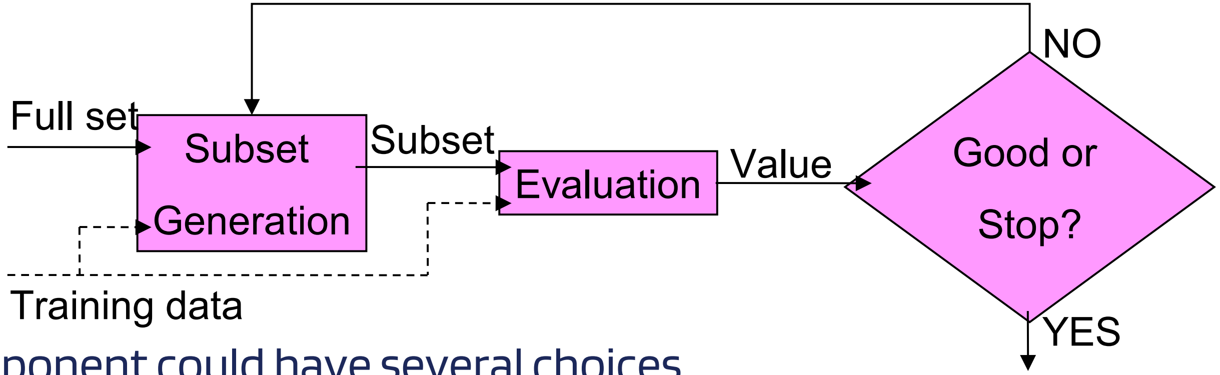
Wrapper *vs.* Filter

- **Wrapper models** try to solve a specific problem, hence the criterion can be really optimized. However, they are potentially very time consuming since they typically need to evaluate a cross-validation scheme at every iteration.
- **Filter models** are much faster but they do not incorporate the data mining algorithm used for the generation of a model or a pattern. Therefore, the result can be suboptimal.

Feature selection

Both the wrapper model and the filter model share a subset selection strategy and a performance evaluation. Indeed it is possible to have a unifying view of the two models.

If we consider the performance estimated by a trained model as another measure, we can unify the filter and wrapper models into a general model.



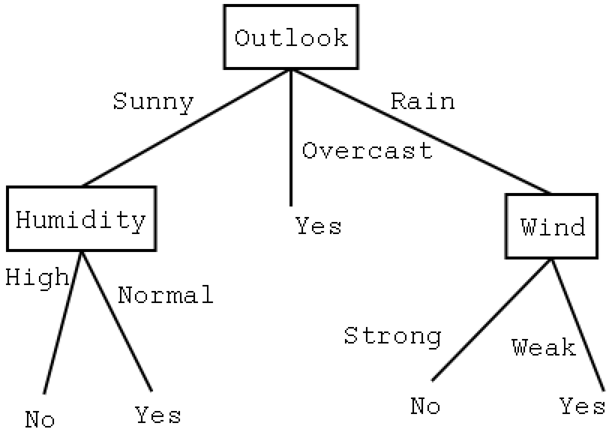
Each component could have several choices.

The various combinations of these choices are the basis of many existing feature selection algorithms.

Select Data

Feature selection

- In contrast to filter and wrapper models, in **embedded methods** the feature selection step cannot be separated from the data mining algorithm. It is an integral part.
- For instance, in decision tree learning, the selection of the features which contribute to the final tree is part of the decision tree construction algorithm.



Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Select Data

Feature selection

- Since we are interested in selecting features in the data transformation step, here we do not consider embedded methods.

Data Preparation

Clean Data

According to the output of data quality verification, data cleaning raises data quality to the required level.

This involves selection of clean data subsets, insertion of suitable defaults or more ambitious techniques such as estimation of missing data modeling.

OUTPUT:

- a data cleaning report, which describes decisions and actions to address data quality problems and lists data transformations for cleaning and possible impacts on the result analysis.

Data Preparation

Clean Data

Three of the most common issues are

- noisy data,
- outliers or anomalies,
- missing values.

Data Preparation

Clean Data

Noisy data. With noisy data, the observed values of one or more variables are different from the actual values. This is due to:

- errors in the observation process

- latent factors which affect the observed variables.

- Human error

  Examples: age = 650; negative income

  Clearly, these values have to be either corrected (if a valid value or reasonable substitute can be found) or dropped from the analysis

Data Preparation

Clean Data

Outliers or anomalies. Values which may occur but are very rare.

- Skewed distributions often indicate outliers.

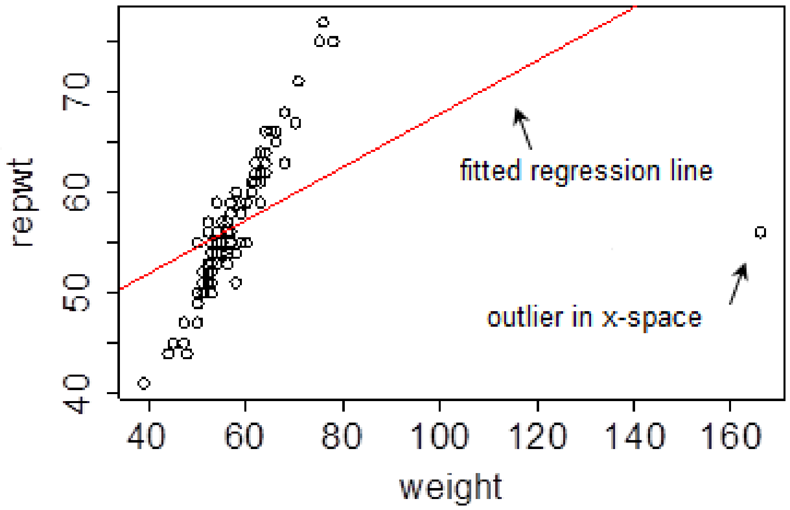
  Examples: A histogram may show that most of the people in a target group have low incomes and only a few are high earners.

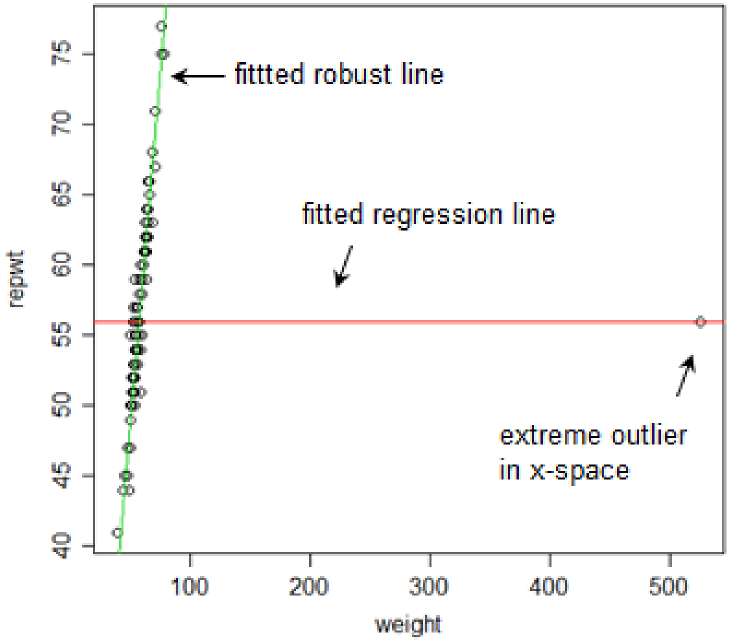
Genuine high earners in this homogeneous group -> Outliers

Poor data collection -> Noise

Data Preparation

Clean Data





Data Preparation

Clean Data

Missing values. Include values that are simply not present in the selected data and those invalid values that we may have deleted during noise detection.

Values may be missing because of

- human error
- the information was not available at the time of input
- the data was selected across heterogeneous sources, thus creating mismatches.

To deal with missing values, data analysts use different techniques, none of which is ideal.

Data Preparation

Clean Data

One technique: *eliminate the observations that have missing values*.

10	BA	1.0	A	0.2
15	TA	1.1	A	0.2
16	BR	?	A	0.3
?	BR	1.6	A	0.1
11	BA	1.0	B	0.3
16	TA	0.8	B	0.1
22	BR	0.9	?	0.4
24	BR	2.0	A	0.5

Drop these rows

Easy to do, but with the obvious drawback of losing valuable data.

Data Preparation

Clean Data

Another technique: *eliminate the variable from the analysis* if there are a significant number of observations with missing values for that same variable.

10	BA	1.0	A	0.2
15	TA	1.1	A	0.2
16	BR	?	A	0.3
12	BR	1.6	A	0.1
11	BA	?	B	0.3
16	TA	0.8	B	0.1
22	BR	?	?	0.4
24	BR	2.0	A	0.5

Data Preparation

Clean Data

Another technique: *replace the missing value with its most likely value*.

Quantitative variable -> mean or median

Categorical variable -> mode or new "unknown" value

A more sophisticated approach: *use a predictive model* to predict the most likely value for a variable on the basis of the values of the other variables in the observation.

Data Preparation

Construct Data

This task includes constructive data preparation operations, such as generation of derived variables, entire new records, or transformed values of existing variables.

OUTPUT:

- derived variables, generated records, and single-variable transformed data

Data Preparation

Construct Data

Single-variable trasformed data. Data is typically further refined to suit the input format requirements of the particular data mining algorithms to be used.

Examples:

- the conversion of date variables from standard US or European format to Julian format
- the computation of an age from a date of birth.
- the aggregation of the average account balance for the last 3-, 6-, and 12-month periods.

Data Preparation

Construct Data

- Scaling or normalization

Normalization is a process where numeric columns are transformed using a mathematical function to a new range. It is important for two reasons.

1. Any analysis of the data should treat all variables equally so that one column doesn't have more influence over another because the ranges are different.
2. Some data mining can accept only numeric input in the range [0.0, 1.0] or [−1.0, +1.0]. In these cases, continuous parameter values must be scaled or normalized to fall within a small, specified range

[min, max]

Data Preparation

Construct Data

min-max normalization

It performs a linear transformation on the original data. Suppose that min_A and max_A are the minimum and maximum values of an attribute A. Min-max normalization maps a value v of A to v' in the range [new_min_A; new_max_A] by computing:

$$v' = \frac{v - \min_A}{\max_A - \min_A}(new\_\max_A - new\_\min_A) + new\_\min_A$$

The relationships among the original data values is preserved. If a future input case for normalization falls outside of the original data range for A, we have an "out of bounds" error.

Normalization to [0,1] is a common form of scaling.

Data Preparation

Construct Data

z-score normalization (also called zero-mean normalization or standardization)

The values for an attribute A are normalized based on the mean and standard deviation of A. A value v of A is normalized to v' by computing:

$$v' = \frac{v - mean_A}{\text{stand\_dev}_\text{A}}$$

where $mean_A$ and $stand\_dev_A$ are the mean and standard deviation, respectively, of attribute A. This method of normalization is useful when the actual minimum and maximum of attribute A are unknown, or when there are outliers which dominate the min-max normalization.

Data Preparation

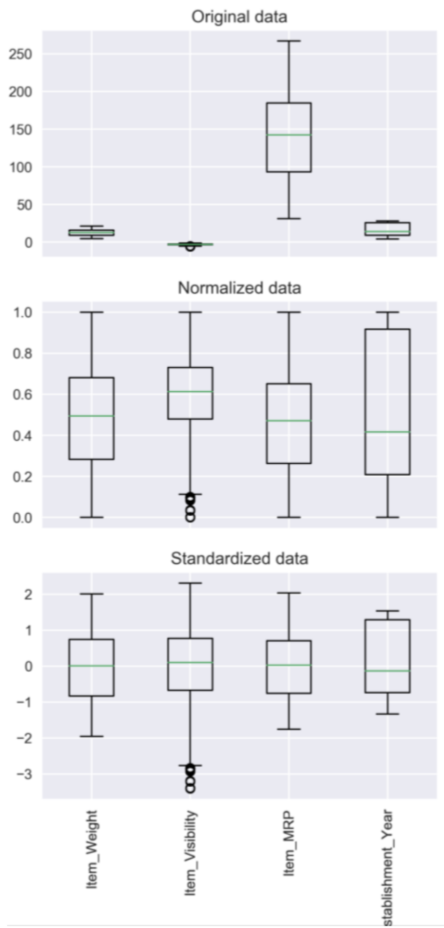
Construct Data

Normalization vs. Standardization

Normalization typically means rescales the values into a range of [0,1]. Standardization typically means rescales data to have a mean of 0 and a standard deviation of 1 (unit variance).

vs.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \qquad X' = \frac{X - \mu}{\sigma}$$



Data Preparation

Construct Data

decimal scaling

This transformation moves the decimal point of values of attribute *A*, to ensure the range is between −1 and +1. The number of decimal points moved depends on the maximum absolute value of *A*. A value *v* of *A* is normalized to *v*′ by computing

$$v' = \frac{v}{10^j}$$

where *j* is the smallest integer such that $max(|v'|) < 1$.

Data Preparation

Construct Data

- Discretization: converting quantitative variables into categorical variables by dividing the values of the input variable into intervals (*bins*).

Two simple discretization (or binning) methods are:

– Equal width

– Equal depth

In both cases, class information is not used in case observations are pre-classified.

Discretization is performed for each attribute independently of the other.

Data Preparation

Construct Data

Equal-width (distance) partitioning:

It divides the range into N intervals of equal size (uniform grid). If A and B are the lowest and highest values of the attribute, the width of intervals will be:

$$W = (B-A)/N.$$

- The most straightforward
- Skewed data is not handled well.
- It is sensitive to outliers

Example: attribute values: 0, 4, 12, 16, 16, 18, 24, 26, 28

Bins:

$[-\infty,10)$	0, 4
$[10,20)$	12, 16, 16, 18
$[20,+\infty)$	24, 26, 28

Data Preparation

Construct Data

Equal-depth (frequency) partitioning:

It divides the range into *N* intervals, each containing approximately the same number of samples.

- Good data scaling
- Interpreting categorical attributes can be tricky.
- Minimizes the information loss during the partitioning process.

Example: attribute values: 0, 4, 12, 16, 16, 18, 24, 26, 28

Bins:

[-∞,14)	0, 4, 12
[14,21)	16, 16, 18
[21,+ ∞)	24, 26, 28

Data Preparation

Construct Data

In both cases the number of bins is defined by the user. Some rules of thumb:

1. In case of classified data, it should not be smaller than the number of classes one wants to recognize
2. Set the number of bins as follows:

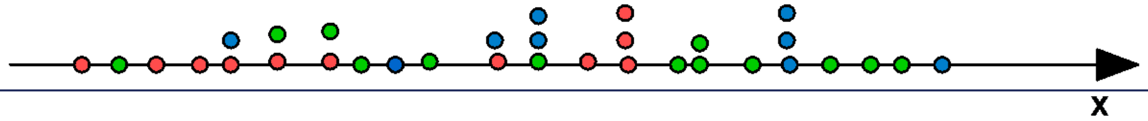
$$N_{bins} = M / (3*C)$$

where:

M – number of training examples

C – number of classes

<u>Example</u>: c = 3 (green, blue, red), M=33 ➔ N_{bins} = 33 / 9 = 4



Data Preparation

Construct Data

- One-of-N converting a categorical variable to a numeric representation, typically for input to a neural network. A special case: One-hot Vector: a binary vector with just one bit equal to 1.

Example:

The categorical variable type of car takes three values, *Ford*, *Lincoln*, and *Nissan*. These values can be represented by transformed values 100, 010, and 001, respectively.

Data Preparation

Construct Data

Complex data conversion procedures aim at building a small set of indices from a large number of variables, such that only some information is lost and the total number of features is reduced.

Example:

FTSE MIB index of the Stock Exchanges in Milan.

Factor analysis and Principal Component Analysis are two statistical *multivariate* analysis techniques for data reduction. We focus on the former.

Data Preparation

Construct Data

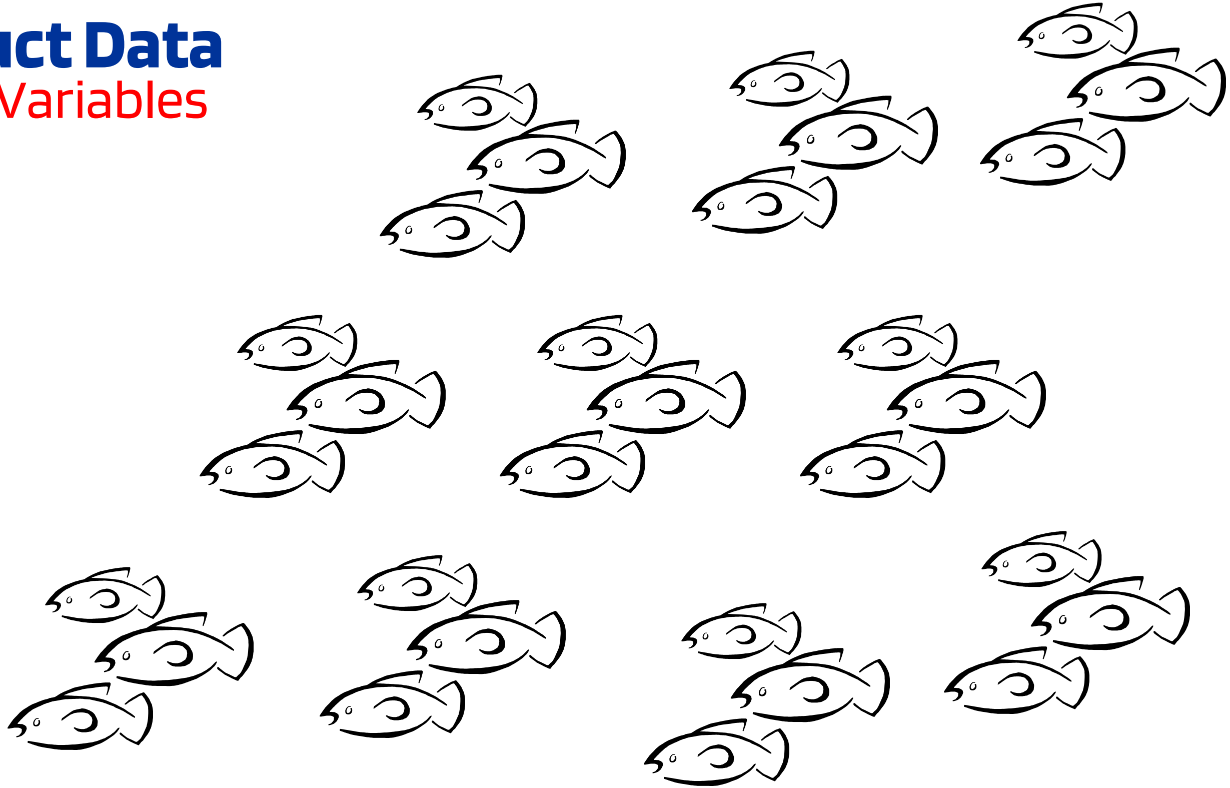
Factor Analysis addresses the problem of analyzing the structure of the interrelationships (correlations) among a LARGE number of variables (e.g., test scores, test items, questionnaire responses) by defining a set of common underlying dimensions, known as factors.

- Not a dependence technique (i.e., multiple regression, discriminant analysis) where one or more variables are explicitly considered the criterion or dependent variable and all others are the predictor or independent variables.
- An interdependence technique in which all variables are simultaneously considered, each related to the others.

Data Preparation

Construct Data

Example: Variables



Data Preparation

Construct Data

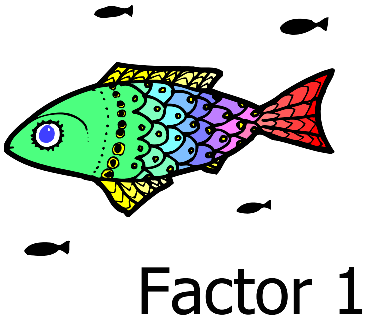
Example: Variables are self-descriptions

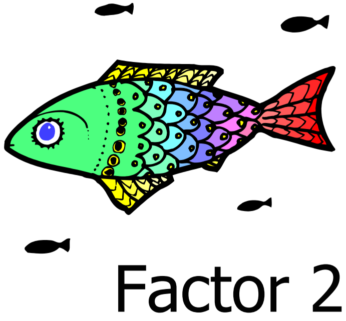
kind	honest
intelligent	sensitive
optimistic	generous
jealous	aggressive
attractive	influential
shy	stubborn
competitive	creative
loving	active
humorous	intense
persistent	lazy

Data Preparation

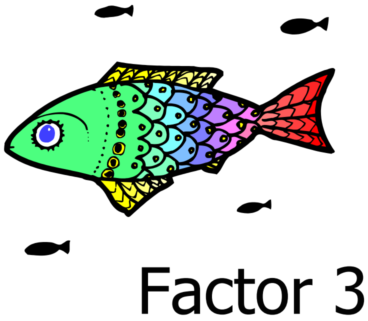
Construct Data

Example: Factors





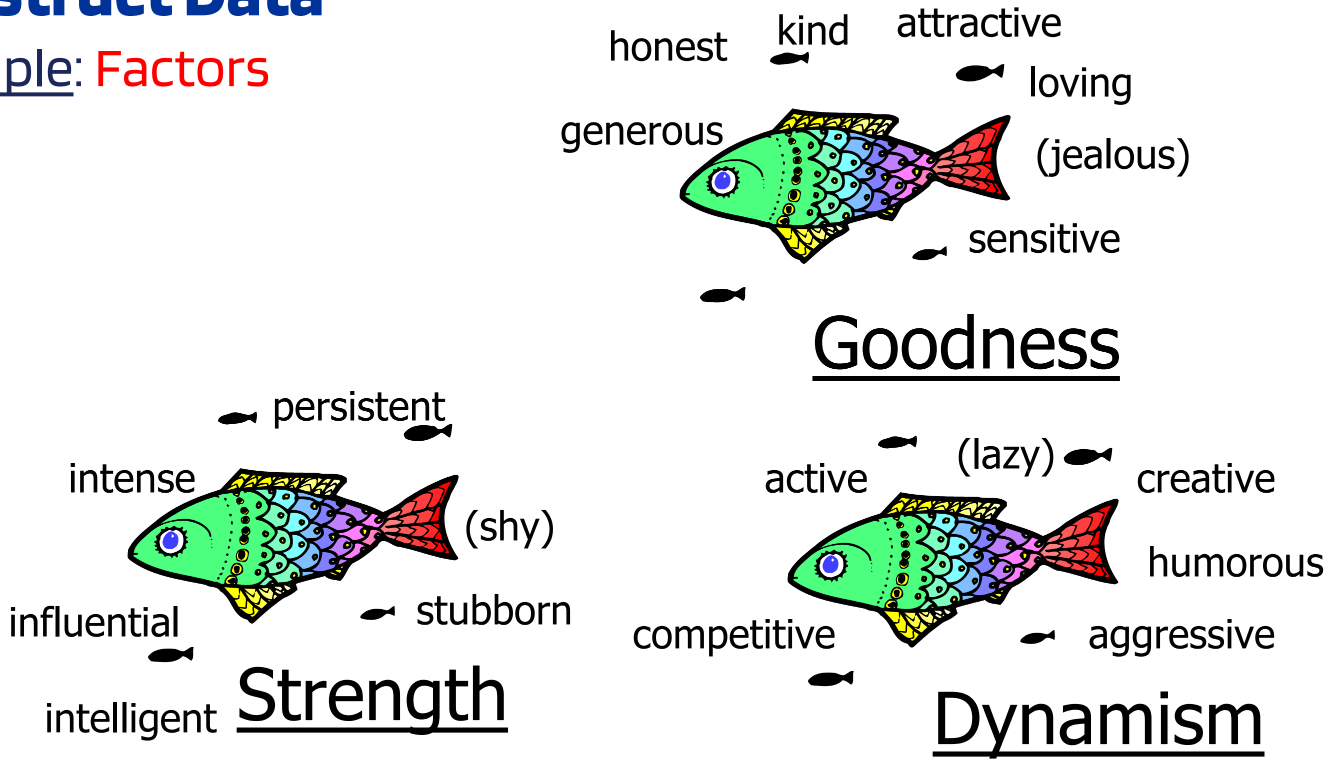
Each factor represents a cluster of variables that correlate closely with one another



Data Preparation

Construct Data

Example: Factors



Data Preparation

Construct Data

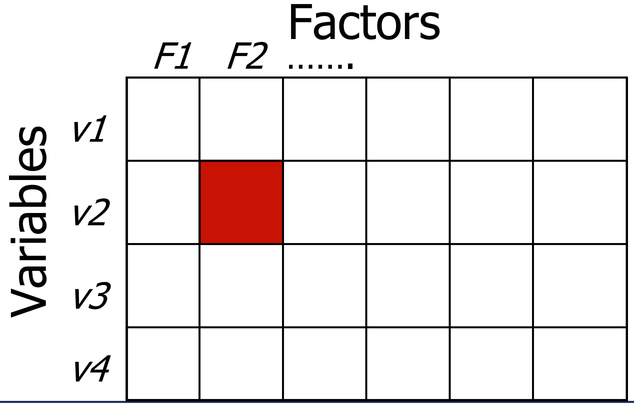
Factors are fewer than original variables.

They can be used for further analysis (data reduction) !!

Algebraic form of a factor:

Y = X1 (variable 1) + X2 (variable 2) + X3 (variable 3)….

where X1, X2, and X3 = factor loadings.



Factor loadings ranges from -1.00 to +1.00. They represent the degree to which each variable correlates with each factor.

Data Preparation

Construct Data

- Factor Loading - can be positive or negative and high or low
- Factor Scores (calculated for original units of observation as a weighted combination of the scores on each input variable)

Y = X1 (variable 1) + X2 (variable 2) + X3 (variable 3)….

- High factor loadings mean high weights in equation for calculating factor scores

Data Preparation

Construct Data

Rule of thumb used by factor analysts

- factor loadings greater than +/- .30 are considered to meet the minimal level;
- loadings of +/- .40 are considered more important;
- if the loadings are +/- .50 or greater, they are considered practically significant.

GOODNESS	*Factor Loadings*
▪ Kind	.71
▪ Generous	.66
▪ Jealous	-.51
▪ Sensitive	.31

Data Preparation

Construct Data

Example in marketing research

A study was conducted by a bank to determine whether or not special marketing programs should be developed for several key segments. One of the study's research questions concerned attitudes toward banking. The respondents were asked to express their opinions on a 0–9, agree–disagree scale …

1. Small banks charge less than large banks (X1)
2. Large banks are more likely to make mistakes than small banks (X2)
3. Tellers do not need to be extremely courteous and friendly, it's enough for them simply to be civil (X3)
4. I want to be known personally at my bank and be treated with special courtesy (X4)
5. If a financial institution treated me in a impersonal or uncaring way, I would never patronize that organization again (X5)

Data Preparation

Construct Data

Example A sample of 15 respondents

Obs	X1	X2	X3	X4	X5
1	9	6	9	2	2
2	4	6	2	6	7
3	0	0	5	0	0
4	2	2	0	9	9
5	6	9	8	3	3
6	3	8	5	4	7
7	4	5	6	3	6
8	8	6	8	2	2
9	4	4	0	8	8
10	2	8	4	5	7
11	1	2	6	0	0
12	6	9	7	3	5
13	6	7	1	7	8
14	2	1	7	1	1
15	9	7	9	2	1

Data Preparation

Construct Data

Example:

Is it possible to represent each of these variables as a linear combination of a *smaller* set of factors? ➔ *factor analysis*

It is possible to group the individuals based on similarity of responses to questions X1 through X5? ➔ *cluster analysis*

Do not confuse clustering of variables (factor analysis) with clustering of observations (cluster analysis)

Data Preparation

Construct Data

Reducing the number of input variables has the following:

Advantage:

- it produces a smaller and more manageable set for further analysis.

Disadvantages:

- difficult task;
- combining variables will cause some loss of information;
- the final result is more difficult to interpret.

Data Preparation

Integrate Data

The integration task aims at combining information from multiple datasets or tables or records to create new records or values.

OUTPUT:

- merged data, which refers to joining two or more dataset/tables that contain different information about the same objects, or aggregated data, which involves operations to compute new values by summarizing information from multiple tables or records.

Data Preparation

Integrate Data

The result is the ***analytical data model***, which represents a consolidated, integrated, and time-dependent restructuring of the data selected and preprocessed from the various operational and external sources.

An important decision made in this task is about the unit of analysis. This is the major **entity** that is being analyzed in the study.

In social science research, the most typical units of analysis are individual people. Other units of analysis can be groups, artifacts (books, photos, newspapers), geographical units (town, census tract, state), social interactions (dyadic relations, divorces, arrests).

Data Preparation

Integrate Data

The unit of analysis should not be confused with the **units of observation**, which is the unit on which data are collected, i.e., systematic observations are made.

For instance, a study may have a unit of observation at the individual level (e.g., pupils), but may have the unit of analysis at the level of a group (e.g., a class).

Example: comparing the average grade of classes.

Example:

- Subject's education – *person* is unit of observation.
- Average education of group to which subject belongs – *group* is level of analysis.

Data Preparation

Integrate Data

In the simplest cases, there is only one unit of observation that coincides with the unit of analysis: the generalizations made from a statistical analysis are attributed to the unit of observation (i.e., the objects on which data were collected and organized for statistical analysis).

Example:

- Analysis of correlation between gender and age.

The conclusion "women have a longer life expectancy than men" is a generalization that applies to the observation unit "Person".

Data Preparation

Integrate Data

A common situation is when the unit of analysis are obtained by aggregating the unit of observations.

Units of <u>observation</u> $\xrightarrow{\textit{aggregation}}$ Units of <u>analysis</u>

Units of observation:

- John's height
- Al's height
- Jody's height
- etc.

Units of analysis:

- Average height of everyone in this group.

Data Preparation

Integrate Data

Obviously, the reverse …

disaggregation

Units of observation → Units of analysis

Units of analysis

- John's height
- Al's height
- Jody's height
- etc.

Units of observation

- Average height of everyone in this group.

can't be done!

Unfortunately, this is not always well understood, and people incur in the typical error of …

Data Preparation

Integrate Data

"**Ecological inference**" (or "*ecological fallacy*") = inferences about individual behavior drawn from data about aggregates.

Examples:

Fact: Towns with more poor have more guns. → Invalid conclusion: The poor are more likely than others to own guns.

Data Preparation

Integrate Data

Examples:

Fact: Protestant countries have higher suicide rates than Catholic countries. → Invalid conclusion: Protestants commit more suicides.

Fact: The more Protestants in a community, the greater the proportion of people who joined the Nazi party in the early 1930s (Germany). → Invalid conclusion: Protestants were especially susceptible to the party's appeals.

Data Preparation

Integrate Data

Examples:

Fact: Protestant countries have higher suicide rates than Catholic countries. → Invalid conclusion: Protestants commit more suicides.

Fact: The more Protestants in a community, the greater the proportion of people who joined the Nazi party in the early 1930s (Germany). → Invalid conclusion: Protestants were especially susceptible to the party's appeals.

If you want to draw conclusions about persons, person should be the unit of analysis!

Data Preparation

Integrate Data

More in general, units of analysis can be composed of several units of observations of different type.

Example: A major national study uses a form that collects information about each person in a dwelling and information about the housing structure. Therefore, this study collects data for two units of observation: persons and housing structures.

From these data, different units of analysis may be constructed:

– Household could be examined as a unit of analysis by combining data from people living in the same dwelling.

– Family could be treated as the unit of analysis by combining data from all members in a dwelling sharing a familial relationship.

Data Preparation

Integrate Data

Many **C**ustomer **R**elationship **M**anagement (CRM) applications use a data transformation technique called householding.

Housolding = consolidating the (typically) dispersed customer records to build of an organization to build a profile of its relationship with a particular household.

This consolidated profile includes entries for all the household members along with details of each of the services and/or products of the organization they use.

Data Preparation

Integrate Data

Example: Two database tables

ID	Name	Address	Age	Life Insurance
AW1	John Smith	6 Oak Str., NY	38	Serene-Future

ID	Name	Address	Age	Car Insurance
1006	Jane Smith	6 Oak Str., NY	35	Safe-Driving

ID	Name	Household	Age	Insurance
1006	Jane Smith	125	35	Car
AW1	John Smith	125	38	Life

Data Preparation

Integrate Data

In all examples considered so far, data of units of analysis are represented in a single two-way table (or in a database relation), where rows describe the units of analysis and the columns represent properties (features) of the units of analysis.

Most existing data mining algorithms operate on an analytical data model with this simple structure (***single table assumption***).

Example: build classification models of customers.

ID	*Name*	*First Name*	*Street*	*City*	*Sex*	*Social Status*	*Income*	*Age*	*Response*
3478	Smith	John	38 Lake St	Seattle	M	single	160k	32	Y
3479	Doe	Jane	45 Sea St	Venice	F	married	180k	45	N
…	…	…	…	…	…	…	…	…	…

Data Preparation

Integrate Data

Example (cont.ed):

- It is possible to add as many attributes about our customers as we like.
  - *A person's number of children*
- For other kinds of information the single-table assumption turns out to be a significant limitation
  - *Add information about orders placed by a customer, in particular*
    - *Delivery and payment modes*
    - *With which kind of store the order was placed (size, ownership, location)*

Add attributes with this information ...

ID	*Name*	*First Name*	*…*	*Respons e*	*Delivery mode*	*Payment mode*	*Store size*	*Store type*	*Locatio n*
3478	Smith	John	…	Y	regular	cash	small	franchis	city
3479	Doe	Jane	…	N	express	credit	large	indep	rural
…	…	…	…	…	…	…	…	…	…

Data Preparation

Integrate Data

However, this solution works fine for once-only customers. What if our business has repeat customers?

Under the single-table assumption we can

1. Make one entry for each order in our customer table

ID	*Name*	*First Name*	*…*	*Response*	*Delivery mode*	*Payment mode*	*Store size*	*Store type*	*Location*
3478	Smith	John	…	Y	regular	cash	small	franchis	city
3478	Smith	John	…	Y	express	check	small	franchis	city
…	…	…	…	…	…	…	…	…	…

We have usual problems of non-normalized tables (redundancies, anomalies, etc.), but the most serious pitfall is that we have one row per order. This means that the analysis results will really be about orders, not customers, which is not what we might want!

Data Preparation

Integrate Data

Alternatively, we can:

2. Aggregate order data into a single tuple per customer.

ID	*Name*	*First Name*	...	*Response*	*No. of orders*	*No. of stores*
3478	Smith	John	…	Y	3	2
3479	Doe	Jane	…	N	2	2
…	…	…	…	…	…	…

No redundancy. Standard DM methods work fine, but

- There is much less information in the new table
- Interactions between the payment mode and the store type are lost

Data Preparation

Integrate Data

A database designer would represent the information in our problem as a set of tables, thus violating the single-table assumption underlying many data mining algorithms (called ***propositional***).

ID	Name	First Name	Street	City	Sex	Social Status	Income	Age	Response
3478	Smith	John	38 Lake St	Seattle	M	single	160k	32	Y
3479	Doe	Jane	45 Sea St	Venice	F	married	180k	45	N
…	…	…	…	…	…	…	…	…	…

Cust ID	Order ID	Store ID	Delivery mode	Payment mode
3478	213444	12	regular	cash
3478	372347	19	regular	cash
3478	334555	12	express	check
…	…	…	…	…

Store ID	size	Type	Location
12	small	franchis	city
19	large	indep	rural
…	…		…

Data Preparation

Integrate Data

- (Multi-)Relational data mining algorithms can analyze data distributed in multiple relations, as they are available in relational database systems.
- These algorithms come from the field of *inductive logic programming* (**ILP**)
  - ILP has been concerned with finding patterns expressed as logic programs
  - Initially, ILP focussed on automated program synthesis from examples
  - In recent years, the scope of ILP has broadened to cover the whole spectrum of data mining tasks (association rules, regression, clustering, …)

Data Preparation
Integrate Data

- Relational data mining algorithms discovers ***relational patterns*** which involve multiple relations.
- They are typically stated in a more expressive language than ***propositional patterns*** defined on a single data table.
- <u>Examples</u> of languages: predicate calculus (a kind of first-order logic), SQL, graphs,
- Different forms of relational patterns, most of which are upgrading of propositional patterns:
  - Relational classification rules
  - Relational regression trees
  - Relational association rules

Data Preparation

Integrate Data

An example of relational classification rule:

IF Customer(C1,N1,FN1,Str1,City1,Zip1,Sex1,SoSt1, In1,Age1,Resp1)
AND order(C1,O1,S1,Deliv1, Pay1)
AND Pay1 = credit_card
AND In1 ≥ 108000
THEN Resp1 = Yes

In SQL:

SELECT C.ID, “Yes”
FROM Customer C, Order O
WHERE C.ID = O.CustID AND O.Payment_mode = ‘credit card’ AND C.
Income >= 108000

Data Preparation
Format Data

Formatting mainly covers syntactic modifications that do not change the meanings, but are required by the modelling tool.

OUTPUT:

- reordered records, rearranged attributes

No industrial standard on formatted data.
For data files the most common standards are:

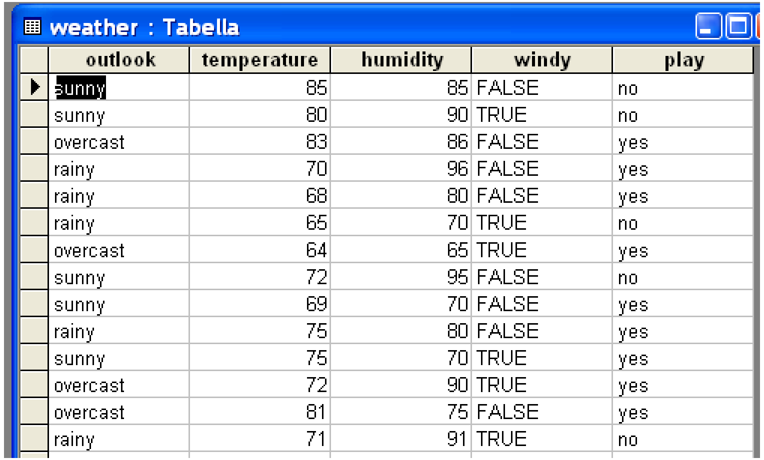
Data Preparation

Format Data

CSV (Comma Separated Value)

Most spreadsheet and database programs allow you to export data into a file in this format as a list of records with commas between items.

```
outlook,temperature,humidity,windy,play
sunny,85,85,FALSE,no
sunny,80,90,TRUE,no
overcast,83,86,FALSE,yes
rainy,70,96,FALSE,yes
rainy,68,80,FALSE,yes
rainy,65,70,TRUE,no
overcast,64,65,TRUE,yes
sunny,72,95,FALSE,no
sunny,69,70,FALSE,yes
rainy,75,80,FALSE,yes
sunny,75,70,TRUE,yes
overcast,72,90,TRUE,yes
overcast,81,75,FALSE,yes
rainy,71,91,TRUE,no
```


weather : Tabella

outlook	temperature	humidity	windy	play
sunny	85	85	FALSE	no
sunny	80	90	TRUE	no
overcast	83	86	FALSE	yes
rainy	70	96	FALSE	yes
rainy	68	80	FALSE	yes
rainy	65	70	TRUE	no
overcast	64	65	TRUE	yes
sunny	72	95	FALSE	no
sunny	69	70	FALSE	yes
rainy	75	80	FALSE	yes
sunny	75	70	TRUE	yes
overcast	72	90	TRUE	yes
overcast	81	75	FALSE	yes
rainy	71	91	TRUE	no

Data Preparation

Format Data

ARFF (Attribute-Relation File Format): proprietary of Weka

```
%
% ARFF file for weather data with some numeric features
%
@relation weather

@attribute outlook {sunny, overcast, rainy}
@attribute temperature numeric
@attribute humidity numeric
@attribute windy {true, false}
@attribute play? {yes, no}

@data
sunny, 85, 85, false, no
sunny, 80, 90, true, no
overcast, 83, 86, false, yes
...
```

weather : Tabella

outlook	temperature	humidity	windy	play
sunny	85	85	FALSE	no
sunny	80	90	TRUE	no
overcast	83	86	FALSE	yes
rainy	70	96	FALSE	yes
rainy	68	80	FALSE	yes
rainy	65	70	TRUE	no
overcast	64	65	TRUE	yes
sunny	72	95	FALSE	no
sunny	69	70	FALSE	yes
rainy	75	80	FALSE	yes
sunny	75	70	TRUE	yes
overcast	72	90	TRUE	yes
overcast	81	75	FALSE	yes
rainy	71	91	TRUE	no

Data Preparation

Format Data

- In some applications most attribute values in a dataset are zero

  Example: word counts in a text categorization problem

- ARFF supports sparse data

```
0, 26, 0, 0, 0 ,0, 63, 0, 0, 0, “class A”
0, 0, 0, 42, 0, 0,  0, 0, 0, 0, “class B”
```

Can be represented as follows:

```
{1 26, 6 63, 10 “class A”}
{3 42, 10 “class B”}
```

Each instance is enclosed in curly braces and contains the index number of each nonzero attribute (indexes start from 0) and its value. Omitted values have a value of 0—they are not “missing” values. If a value is unknown, it must be explicitly represented with a question mark (?).

Data Preparation
Format Data

- ARFF supports string attributes:

```
@attribute description string
```

Similar to nominal attributes but list of values is not pre-specified

- It also supports date attributes:

```
@attribute today date
```

Uses the ISO-8601 combined date and time format yyyy-MM-dd-THH:mm:ss

- String and date attributes are actually nominal and numeric, respectively.

Customer Vulnerability Analysis (CVA)

The team summarized the three-year history of data for each customer into a single record by creating 20 new variables, including

- ✓ *Primary- Brand* (the orange juice brand of which the household bought the most ounces)
- ✓ *Purchase-Frequency* (the ratio of the number of purchases to the number of days between the first and the last supermarket visit)
- ✓ *Ounces-Per-Day* (the ratio of the number of ounces of orange juice purchased to the number of days between the first and the last supermarket visit).

The unit of analysis is the customer.

Customer Vulnerability Analysis (CVA)

The team had to define the concepts of vulnerable customer and loyal customer. It was decided that any consumer who purchased the same brand of orange juice more than 80% of the time would be considered loyal.

All others where labelled as “vulnerable”.

The resulting database, called the OJ database, had 1,986 records, and each record was labelled as either loyal or vulnerable.